

MALLA REDDY UNIVERSITY

IV Year B. Tech – I Semester

L/T/P/C
0/0/2/1**(MR20-1CS0383) BIG DATA ANALYTICS LAB****COURSE OBJECTIVES:**

1. Provide the knowledge to setup a Hadoop Cluster.
2. Impart knowledge to develop programs using MapReduce.
3. To install and configure PySpark on local and cluster environments..
4. Present latest big data frameworks and applications using PySpark
5. Perform basic data analytics using PySpark functions to read data from CSV files into PySpark Data Frames.

TASK-1: Installation of Hadoop on Single Node Setup**TASK-2:** Understanding and using basic HDFS commands**TASK-3:** Run a basic word count Map Reduce program to understand Map Reduce Paradigm.**TASK-4:** Write a Map Reduce program that mines weather data.**TASK-5:** Install and configure PySpark on a local system, enabling the development and execution of Apache Spark applications using the Python programming language.**TASK-6:** Perform simple data transformation like filtering even numbers from a given list using PySpark RDD.**TASK-7:** Explore how the `collect()` operation works in PySpark using a dataset with basic RDD operations."**TASK-8:** Create a DataFrame in PySpark by reading data from a CSV file and explore its structure and contents.**TASK-9:** Create a DataFrame in PySpark and apply basic operations such as viewing data and selecting columns."**TASK-10:** Implement and demonstrate dataset sampling using the `sample()` and `takeSample()` methods in PySpark.(DataFrames)**TASK-11:** Create a DataFrame in PySpark by loading data from a CSV file and perform basic analytical operations

REFERENCE BOOKS:

1. Tom White, “Hadoop: The Definitive Guide”, 4th Edition, O’Reilly Inc, 2015.
2. Tanmay Deshpande, “Hadoop Real-World Solutions Cookbook”, 2nd Edition, Packt Publishing, 2016.
3. Edward Capriolo, Dean Wampler, and Jason Rutherglen, “Programming Hive”, O’ReillyInc, 2012.
4. Vignesh Prajapati, “Big data Analytics with R and Hadoop”, Packt Publishing, 2013.

COURSE OUTCOMES:

1. Understand Hadoop working environment.
2. Apply Map Reduce programs for real world problems.
3. Implement scripts using Pig to solve real world problems.
4. Analyze queries using Hive to analyze the datasets
5. Understand spark working environment and integration with R